# 12th HACKERS KA JUGAAD

# Atoms

## Thomson's Model of an Atom

- In 1898 JJ Thomson proposed that an atom is a sphere of positively charged matter with electrons embedded in it
- The positive charge is uniformly distributed over the entire atom. The arrangement of electrons inside the continuous positive charge is similar to that of the seeds in the watermelon or the plums is in a pudding.
- That is why the Thomson atomic model is also known as plum pudding model
- The electrons are arranged in such a manner that their mutual repulsions are balanced by their attractions with the positively charged matter. Does the atom as a whole is stable and neutral.

## failure of Thomson's Model

- It could not explain the origin of several spectral series in the case of hydrogen and other atoms
- It failed to explain the larger angle scattering of alpha particles in Rutherfords experiment

## Rutherford's Alpha - particle scattering

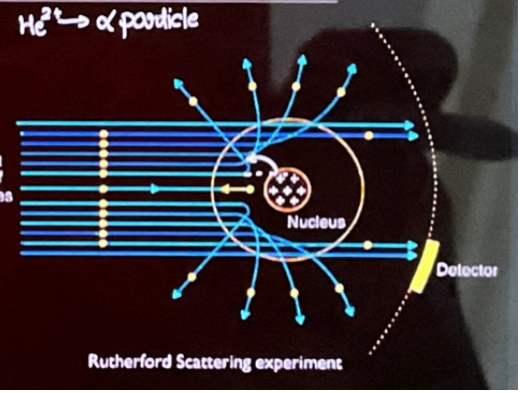
On the suggestion of Rutherford, in 1911, his two associates, **H. Geiger** and **E. Marsden**, performed an experiment **by bombarding α-particles (Helium nuclei Z = 2, A = 4) on a gold foil.**

$He^{2+} \rightarrow \alpha$ particle

**Observations :**

(i) Most of the α -particles pass through the gold foil undeflected.

(ii) A very small number of α-particles (1 in 8000) suffered large angle deflection; some of them retraced their path or suffered 180° deflection.



Rutherford Scattering experiment

**Conclusion:**

(i) Atom is hollow.

(ii) Entire positive charge and nearly whole mass of atom is concentrated in a small center called nucleus of atom.

(iii) Coulomb's law holds good for atomic distances.

(iv) Negatively charged electrons are outside the nucleus.



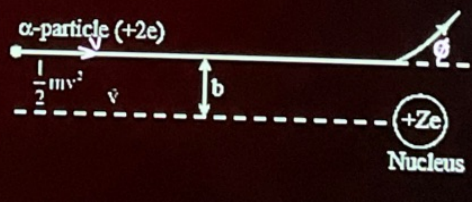
Rutherford Scattering experiment

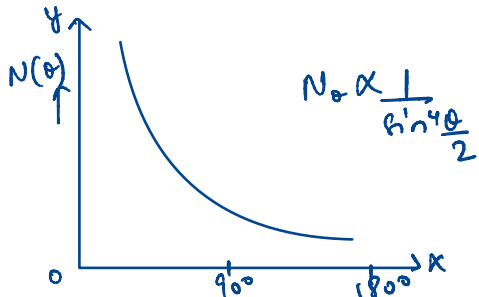
**Impact Parameter:** The perpendicular distance of initial velocity vector of α-particle from the nucleus, when the particle is far away from the nucleus, is called the impact parameter. It is denoted by b. **For head on approach of α-particle, b = 0.**

**Angle of Scattering (ϕ)**: The angle by which α-particle is deviated from its original direction is called angle of scattering.

$$b = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{E_K}\cot\frac{\phi}{2}$$

where $E_k$ is the initial kinetic energy




$N_\theta \propto \frac{1}{\sin^4\frac{\theta}{2}}$

$\frac{N_\theta}{N_{\theta'}} = \frac{\sin^4\theta'/2}{\sin^4\theta/2}$

$\frac{1000}{N_{90°}} = \frac{\sin^4 45°}{\sin^4 30°}$

# Distance of closest approach

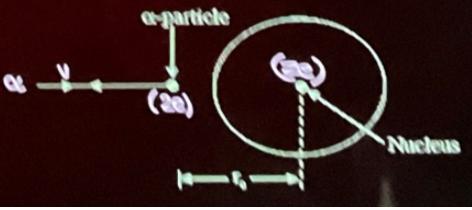
The smallest distance of approach of α-particle near heavy nucleus is a measure of the size of nucleus.

Distance of nearest approach = size of nucleus $= \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{E_K}$

where $E_K$ is kinetic energy of incident α-particle, Z = atomic number, e = electronic charge.

$$\frac{1}{2}mv^2 = E_K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0}$$

$$r_0 = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{E_K}$$




# Rutherford's Atom Model

- Every atom consists of a tiny central core called the nucleus which contains all of the atoms, positive charge and most of its mass. Therefore, alpha particle travelling close to the nucleus will have large scattering angle. This rutherfords model explains large angle scattering of alpha particles.

- The radius of the nucleus is of the order $10^{-15}$ and that of the atom is. $10^{-10}$ of the order. Therefore, the nucleus occupies only a small portion of the available Space

- The electrons occupy the space outside the nucleus. Since an atom is electrically neutral the positive charge of the nucleus is equal to the negative charge on the electrons.

- Electrons are not stationary but revolve around the nucleus in circular path. In this way Rutherford provide stability to the atom. It is because now the force of attraction between the electrons and the nucleus provided the necessary centripetal force to the revolving electrons.

# Limitations of Rutherford's atomic Model

- According to electromagnetic theory an accelerated charged, particle must radiate electromagnetic energy an electron revolving around the nucleus is at the continuous acceleration towards the centre. It should continuously lose energy and move in orbit of gradually decreasing radii. The electrons should follow spiral path and finally it should collapse into the nucleus. Thus the Rutherford's model cannot explain the stability of an atom.

- In Rutherford's model an electron can be revolve in orbit of all possible radii so it should emit a continuous spectrum but an atom like hydrogen always emits a discrete Line spectrum

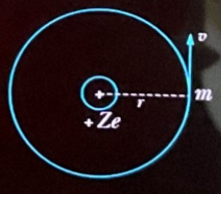
# Bohr's Model

Bohr modified Rutherford atom model to explain the line spectrum of hydrogen.

**Postulates of Bohr's Theory**

**(i) Stationary Circular Orbits:** An atom consists of a central positively charged nucleus and negatively charged electrons revolve around the nucleus in certain orbits called **stationary orbits**. The electrostatic coulomb force between electrons and the nucleus provides the necessary centripetal force.

$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{(Ze)(e)}{r^2} \quad \ldots\ldots\ldots\ldots \text{(i)}$$




where Z is the atomic number,
m is the mass of electrons,
r = radius of orbit.

**(ii) Quantum Condition:** The stationary orbits are those in which angular momentum of electron is an integral multiple of $\frac{h}{2\pi}$, i.e.

$$mvr = n\frac{h}{2\pi} \qquad n = 1, 2, 3, \ldots\ldots\ldots\ldots \text{(ii)}$$

Integer $n$ is called the principal quantum number. This equation is called Bohr's quantum condition.

$h = 6.6 \times 10^{-34}$ Js

$mvn = n\left(\frac{h}{2\pi}\right)$

$n = 3$

**(iii) Transitions:** The **electron does not radiate energy when in a stationary orbit.** The quantum of energy (or photon) is emitted or absorbed when an electron jumps from one stationary orbit to the other. The frequency of emitted or absorbed photon is given by

$$\frac{hc}{\lambda} = \boxed{h\nu = |E_i - E_f|} \quad \text{..... (iii)}$$

The is called Bohr's frequency condition

## Radius of orbit and energy of electron in orbit:

condition of motion of electron in circular orbit is

$$r = \frac{2e^2}{4\pi\varepsilon_0 mv^2} = \frac{nh}{2\pi mv}$$

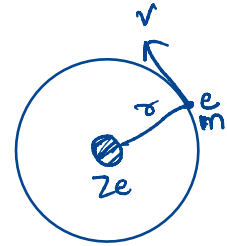
$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(e)}{r^2}$$

Bohr's quantum condition.

$$mvr = n\frac{h}{2\pi}$$

$$v = \frac{2\pi 2e^2}{4\pi\varepsilon_0 nh}$$

$$v = \frac{nh}{2\pi mr}$$




$$v = \frac{e^2}{2\varepsilon_0 h}\cdot\frac{2}{n}$$

Substituting this value of v in (i) we get

$$\frac{m}{r}\left(\frac{nh}{2\pi mr}\right)^2 = \frac{1}{4\pi\varepsilon_0}\frac{2e^2}{r^2}$$

$$v = \left(\frac{c}{137}\right)\frac{z}{n}$$

this gives

| H | $He^+$ | $Li^{2+}$ |
|---|---|---|
| z = 1 | z = 2 | z = 3 |

$$r = \frac{\varepsilon_0 h^2 n^2}{\pi m z e^2}$$

Denoting radius of $n^{th}$ orbit by $r_n$, we have

$$r_n = \frac{\varepsilon_0 h^2 n^2}{\pi m z e^2} \Rightarrow r_n = \left(\frac{\varepsilon_0 h^2}{\pi m e^2}\right)\frac{n^2}{z}$$

$$\Rightarrow r_n = \frac{n^2}{z}(0.529\text{A}^\circ)$$

for hydrogen atom $Z = 1$

$$(r_n)_H = \frac{\varepsilon_0 h^2 n^2}{\pi m e^2}$$

the radius of first orbit of H-atom is called Bohr's radius. It's denoted by $a_0$.

$$\Rightarrow a_0 = \frac{\varepsilon_0 h^2}{\pi m e^2} = 0.529 \times 10^{-10}\ m = 0.529\ \text{Å}$$

**Energy of Orbiting Electron**

From equation (i), $\frac{1}{2}mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2r}$

Kinetic energy, $K = \frac{1}{2}mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2r}$

Potential energy $U = \frac{1}{4\pi\varepsilon_0}\frac{(Ze)(-e)}{r} = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$ — Attraction

Total energy $E = K + U = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2r} - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$

$$\Rightarrow E = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2r}$$

Attractive

$n_1 \rightarrow n_2$ radiation emit

higher lower

$$E_{n_1} - E_{n_2} = h\nu$$

$$\left(\frac{-mZ^2e^4}{8\varepsilon_0^2h^2n_1^2}\right) - \left(\frac{-mZ^2e^4}{8\varepsilon_0^2h^2n_2^2}\right) = \frac{hc}{\lambda}$$

$$\frac{mZ^2e^4}{8\varepsilon_0^2h^2}\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right] = \frac{hc}{\lambda}$$

$$\frac{1}{\lambda} = \frac{mZ^2e^4}{8\varepsilon_0^2h^3c}\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]$$

$$\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]$$

For nth orbit, writing $E_n$ for $E$, we have

$$E_n = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2r_n}$$

Substituting the value of $r_n$ from (ii) in (iv), we get

$$E_n = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2\left(\frac{\varepsilon_0 h^2 n^2}{\pi m Ze^2}\right)} = -\frac{mZ^2e^4}{8\varepsilon_0^2h^2n^2} \Rightarrow E_n = -\frac{\left(\frac{me^4}{8\varepsilon_0^2h^2}\right)}{n^2}Z^2$$

$\downarrow$ 13.6 eV

For convenience introducing Rydberg constant,

$$R = \frac{me^4}{8\varepsilon_0^2ch^3}$$

The value of Rydberg constant is $1.097 \times 10^{7}\ m^{-1}$

$$E_n = -\frac{Z^2}{n^2}(13.6\text{eV})$$

We have

$$E_n = -\frac{Z^2Rhc}{n^2}$$

For hydrogen atom Z = 1,
Energy of orbiting electron in H-atom

$1eV = 1.6 \times 10^{-19} J$

$$E_n = -\frac{Rhc}{n^2}$$

$$\Rightarrow \boxed{E_n = -\frac{13.6}{n^2} eV}$$

Equations (iii) and (vii) indicate that radii and energies of hydrogen like atoms (i.e., atoms containing one electron only) are quantised.

# Energy Levels of Hydrogen Atom

The energy of electron in hydrogen atom (Z = 1) is given (or series of hydrogen spectrum) by $E_n = -\frac{Rhc}{n^2} = -\frac{13.6}{n^2} eV$

When n = 1, $\boxed{E_1 = -13.6 eV}$

When n = 2, $\boxed{E_2 = -\frac{13.6}{4} eV = -3.4 eV}$

When n = 3, $\boxed{E_3 = -\frac{13.6}{9} eV = -1.51 eV}$

When n = 4, $\boxed{E_4 = -\frac{13.6}{16} eV = -0.85 eV}$

When n = 5, $\boxed{E_5 = -\frac{13.6}{25} eV = -0.54 eV}$

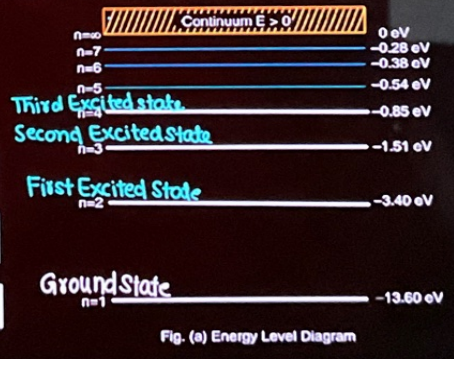



Fig. (a) Energy Level Diagram

When n = 6, $E_6 = -\frac{13.6}{36} eV = -0.38 eV$

When n = 7, $E_7 = -\frac{13.6}{49} eV = -0.28 eV$

.....

.....

When $n = \infty$, $E_\infty = -\frac{13.6}{(\infty)^2} eV = 0 eV$

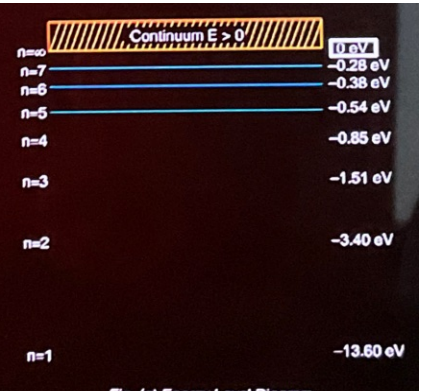



Fig. (a) Energy Level Diagram

# Hydrogen Spectrum

Hydrogen emission spectrum consists of 5 series.
(i) Lyman series: This lies in ultraviolet region.
(ii) Balmer series: This lies in the visible region
(iii) Paschen series: This lies in near infrared region.
(iv) Brackett series: This lies in mid infrared region.
(v) Pfund series: This lies in far infrared region.

Hydrogen absorption spectrum consists of only Lyman series.
Explanation of Hydrogen Spectrum: $n_i$ and $n_f$ are the quantum numbers of initial and final states and E, and E are energies of electron in H-atom (Z=1) in initial and final states then we have

$$E_i = -\frac{Rhc}{n_i^2} \text{ and } E_f = -\frac{Rhc}{n_f^2}$$

Energy of absorbed photon

$$\Delta E = E_f - E_i = Rhc\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \quad \text{.....(ix)}$$

If v is the frequency of emitted radiation, we have from Bohr's fourth postulate

$$\nu = \frac{E_i - E_f}{h} = -\frac{Rc}{n_i^2} - \left(-\frac{Rc}{n_f^2}\right) = Rc\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \quad \text{.....(ix)}$$

The wave number (i.e., reciprocal of wavelength) of the emitted radiation is given by $\bar{\nu} = \frac{1}{\lambda} = \frac{\nu}{c} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$

The relation explains successfully the origin of various lines in the spectrum of hydrogen atom. The series of lines are obtained due to the transition of electron from various other orbits to a fixed inner orbit.




(i) Lyman series: This series is produced when electron jumps from higher orbits to the first stationary orbit (i.e., $n_f$=1). Thus for this series

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n_i^2}\right) \quad \text{where } n_i = 2,3,4,5,\ldots \longrightarrow n_f = 1$$

$\nu = \frac{c}{\lambda}$

For longest wavelength of lymen series $n_i = 2$

$$\therefore \frac{1}{\lambda_{max}} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3R}{4}$$

$$\therefore \lambda_{max} = \frac{4}{3R} = \frac{4}{3\times1.097\times10^7}\ \text{m}$$

(i) Lyman series:

For shortest wavelength of lymen series $n_i = \infty$

$$\therefore \frac{1}{\lambda_{min}} = R\left(\frac{1}{1^2} - \frac{1}{\infty}\right) = R$$

$$\lambda_{min} = \frac{1}{R} = \frac{1}{1.097\times10^7}\ \text{m} = 0.9116 \times 10^{-7}\ \text{m} = 911.6\text{Å}$$

This is called series limit of Lyman series $\lambda$ = 911.6 Å
Obviously the lines of Lyman series are found in ultraviolet region.

**(ii) Balmer series:** The series is produced when an electron jumps from higher orbits to the second stationary orbit ($n_f = 2$) Thus for this series,

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n_i^2}\right)$$ where $n_i = 3,4,5,6, \ldots$

$H_\alpha$ $H_\beta$ $H_\gamma$

For longest wavelength of Balmer series ($n_i = 3$)

$$\frac{1}{\lambda_{max}} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5R}{36}$$

$$\lambda_{max} = \frac{36}{5R} = \frac{36}{5\times1.097\times10^7}\ m = 6.563 \times 10^{-7}\ m = 6563Å$$

$H_\alpha$

For Shortest wavelength (or series limit) of Balmer series $n_i \to \infty$

$$\therefore \quad \frac{1}{\lambda_{min}} = R\left(\frac{1}{2^2} - \frac{1}{\infty}\right) = \frac{R}{4}$$

$$\lambda_{min} = \frac{4}{R} = \frac{4}{1.097\times10^{-7}}\ m = 3.646 \times 10^{-7}\ m = 3646Å$$

Obviously the lines of Balmer series are found in the visible region and first, second, third ... lines are called $H_\alpha$, $H_\beta$, $H_\gamma$,..., lines respectively

**(iii) Paschen series:** This series is produced when an electron jumps from higher orbits to the third stationary orbit ($n_f = 3$)

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{n_i^2}\right)$$ where $n_i = 4,5,6,7, \ldots$

For Longest wavelength of Paschen series ($n_i = 4$)

$$\therefore \quad \frac{1}{\lambda_{max}} = R\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = \frac{7R}{144}$$

$$\therefore \quad \lambda_{max} = \frac{144}{7R} = \frac{144}{7\times1.097\times10^7}\ m = 18.752 \times 10^{-7}\ m = 18752Å$$

**(iv) Brackett series:** This series is produced when an electron jumps from higher orbits to the fourth stationary orbit ($n_f = 4$)

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{n_i^2}\right)$$ where $n_i = 5,6,7,8, \ldots$

**(v) Pfund series:** This series is produced when an electron jumps from higher orbits to the fifth stationary orbit ($n_f = 5$)

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{5^2} - \frac{1}{n_i^2}\right)$$ where $n_i = 6,7,8, \ldots$

The last three series are found in infrared region.

# Limitations of Bohr's theory

- Bohr's model is applicable to simplest atoms like hydrogen with Z=1 it fails for other elements
- This theory does not explain the fine structure of spectral lines in the hydrogen atom
- It could not explain the difference in the densities of emitted radiations
- It does not take into account the wave properties of electrons
- Bohr's model does not explain why the orbits are circular while, elliptical orbits are also possible

THANK You !!